Lecture 3. Conditional Probability and Independence

Multiplication rule, total probability, Bayes's formula, independence, gambler's ruin

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3.1 Conditional probability

Motivation. Conditioning restricts the sample space to what is known, and the multiplication rule builds probabilities step by step.

Definition (Conditional probability)

If $P(F) > 0$, the conditional probability of E given F is $P(E \mid F) = P(EF)/P(F)$.

With equally likely outcomes $P(E \mid F) = |EF|/|F|$: shrink the sample space to F.

Proposition 3.1 (Multiplication rule)

$$P(E_1E_2\cdots E_n) = P(E_1)P(E_2 \mid E_1)P(E_3 \mid E_1E_2)\cdots P(E_n \mid E_1\cdots E_{n-1}).$$

Proof. Write each factor as a ratio; the product telescopes. ■

Example 3.1

(a) 8 red and 4 white balls, two drawn without replacement: $P(\text{both red}) = \frac{8}{12}\cdot\frac{7}{11} = \frac{14}{33} = \binom{8}{2}/\binom{12}{2}$.
(b) A deck is split at random into four piles of 13; E_i: pile i has exactly one ace. Placing the aces one at a time (the second must fall in one of the 39 of the remaining 51 positions in other piles, and so on),

$$P(E_1E_2E_3E_4) = 1\cdot\frac{39}{51}\cdot\frac{26}{50}\cdot\frac{13}{49} \approx 0.105,$$

Example 3.2 (Matching, recursively)

In Example 2.4, let P_N be the probability of no match and M the event that person 1 takes their own hat; $P_N = P(\text{no match} \mid M^c)\frac{N-1}{N}$. Given that person 1 took hat $j \neq 1$, either person j takes hat 1 (probability $\frac{1}{N-1}$) and the other $N-2$ have no match, or, treating hat 1 as j's own, the $N-1$ people have no match. So

$$P_N = \frac{N-1}{N}\Big(P_{N-1} + \frac{P_{N-2}}{N-1}\Big), \qquad \text{i.e.} \qquad P_N - P_{N-1} = -\frac{1}{N}(P_{N-1} - P_{N-2}).$$

With $P_1 = 0$, $P_2 = \frac{1}{2}$: $P_N - P_{N-1} = (-1)^N/N!$, hence $P_N = \sum_{r=0}^{N}(-1)^r/r!$.

3.2 Total probability and Bayes's formula

Motivation. Bayes's formula inverts $P(B \mid A_i)$ into $P(A_i \mid B)$, often counterintuitively.

Since $E = EF \cup EF^c$,

$$P(E) = P(E \mid F)P(F) + P(E \mid F^c)(1 - P(F)),$$

a weighted average. More generally:

Theorem 3.2 (Total probability and Bayes's formula)

If $F_1, \dots, F_n$ are mutually exclusive with union S, then

$$P(E) = \sum_{i=1}^{n} P(E \mid F_i)P(F_i), \qquad P(F_j \mid E) = \frac{P(E \mid F_j)P(F_j)}{\sum_i P(E \mid F_i)P(F_i)}.$$

$P(F_j)$ are **prior** and $P(F_j \mid E)$ **posterior** probabilities. In odds form,

$$\frac{P(H \mid E)}{P(H^c \mid E)} = \frac{P(H)}{P(H^c)} \cdot \frac{P(E \mid H)}{P(E \mid H^c)} :$$

evidence multiplies the odds by the **likelihood ratio**.

Example 3.3 (A screening test)

Sensitivity 0.95, false-positive rate 1%, prevalence 0.5%:

$$P(D \mid +) = \frac{0.95 \cdot 0.005}{0.95 \cdot 0.005 + 0.01 \cdot 0.995} \approx 0.323.$$

Only a third of positives are diseased: the many healthy produce more false positives than there are true ones.

Example 3.4 (Multiple choice)

A student knows the answer with probability p, otherwise guesses among m alternatives:

$$P(\text{knew} \mid \text{correct}) = \frac{p}{p + (1-p)/m} = \frac{mp}{1 + (m-1)p}.$$

With $m = 5$, $p = 1/2$ this is $5/6$.

Example 3.5 (Three cards)

Cards: red–red, black–black, red–black. A random card shows red on top. Not $1/2$: of three equally likely red faces, two belong to the red–red card, so $P(RR \mid R \text{ up}) = \frac{1/3}{1/3 \cdot 1 + 1/3 \cdot 1/2} = \frac{2}{3}$.

Example 3.6 (Searching for a plane)

A plane is equally likely in one of three regions; a search of region i misses it there with probability β_i. After an unsuccessful search of region 1,

$$P(R_1 \mid \text{miss}) = \frac{\beta_1/3}{\beta_1/3 + 1/3 + 1/3} = \frac{\beta_1}{\beta_1 + 2}, \qquad P(R_j \mid \text{miss}) = \frac{1}{\beta_1 + 2} \quad (j = 2, 3).$$

For fixed F, $Q(E) = P(E \mid F)$ is a probability, e.g. $P(E_1 \mid F) = P(E_1 \mid E_2F)P(E_2 \mid F) + P(E_1 \mid E_2^cF)P(E_2^c \mid F)$.

3.3 Independent events

Definition (Independence)

E and F are **independent** if $P(EF) = P(E)P(F)$. Events $E_1, \dots, E_n$ are (mutually) independent if $P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$ for every subcollection.

Independence of E, F implies that of E, F^c: $P(EF^c) = P(E) - P(E)P(F) = P(E)P(F^c)$.

Example 3.7 (Pairwise does not imply mutual)

Two fair dice; E: sum 7, F: first die 4, G: second die 3. Each pair is independent, but $P(E \mid FG) = 1 \neq P(E)$. Also, sum 6 is *not* independent of F, while E is ($P(\text{sum} = 7 \mid \text{first} = k) = 1/6$).

Example 3.8 (Reliability)

n independent components, component i working with probability p_i: a parallel system works with probability $1 - \prod_i (1 - p_i)$, a series system $\prod_i p_i$.

Example 3.9 (E before F)

Independent trials, E and F disjoint on each. Conditioning on the first trial, $P(E \text{ first}) = P(E) + (1 - P(E) - P(F))P(E \text{ first})$, so

$$P(E \text{ occurs before } F) = \frac{P(E)}{P(E) + P(F)}.$$

Craps: a first roll of 7 or 11 wins, 2, 3, 12 loses; otherwise the sum k is the "point" and one rolls until k (win) or 7 (lose). With $p_k = P(\text{sum} = k)$,

$$P(\text{win}) = p_7 + p_{11} + \sum_{k \in \{4,5,6,8,9,10\}} p_k \frac{p_k}{p_k + p_7} = \frac{244}{495} \approx 0.493.$$

Example 3.10 (The problem of the points)

Probability of n successes before m failures in trials with success probability p: at least n successes in the first $m + n - 1$ trials, so

$$P_{n,m} = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k (1-p)^{m+n-1-k}.$$

(Pascal–Fermat: split the stakes of an interrupted game as $P_{n,m} : 1 - P_{n,m}$; e.g. 0.828 for $p = \frac{1}{2}$, $n = 4$, $m = 7$.)

Example 3.11 (Gambler's ruin)

A gambler with fortune i wins or loses one unit per play (probabilities p, $q = 1 - p$) until N or 0; P_i = probability of reaching N. Conditioning on the first play, $P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1})$,

$P_0 = 0$, so $P_{i+1} - P_i = (q/p)^i P_1$; summing and using $P_N = 1$,

$$P_i = \begin{cases} \dfrac{1-(q/p)^i}{1-(q/p)^N}, & p \neq \frac{1}{2}, \\ \dfrac{i}{N}, & p = \frac{1}{2}, \end{cases}$$

As $N \to \infty$: ruin is certain if $p \leq \frac{1}{2}$; if $p > \frac{1}{2}$ survival has probability $1 - (q/p)^i$.

3.4 $P(\cdot \mid F)$ is a probability

So one may condition sequentially; E_1, E_2 are conditionally independent given F if $P(E_1E_2 \mid F) = P(E_1 \mid F)P(E_2 \mid F)$.

Example 3.12 (Laplace's rule of succession)

Coin $i \in \{0, \ldots, k\}$ shows heads with probability i/k; a random coin gives n heads in n flips. Flips are independent given the coin, so

$$P(\text{head on flip } n+1 \mid n \text{ heads}) = \frac{\sum_i (i/k)^{n+1}/(k+1)}{\sum_i (i/k)^n/(k+1)} \approx \frac{\int_0^1 x^{n+1}dx}{\int_0^1 x^n dx} = \frac{n+1}{n+2}$$

for large k. Unconditionally the flips are dependent: early heads indicate a good coin.

Example 3.13 (Updating sequentially)

If evidence $E_1, \ldots, E_k$ is conditionally independent given each hypothesis H_i,

$$P(H_i \mid E_1 \cdots E_k) \propto P(H_i) \prod_{j=1}^{k} P(E_j \mid H_i),$$

so each posterior is the next prior, in any order.

3.5 Selected exercises

Motivation. Ross's theoretical exercises (book numbering), solved by conditioning on a first or last event.

Exercise 3.7(a) (The colour that survives)

An urn contains $n \geq 1$ white and $m \geq 1$ black balls. The balls are withdrawn one at a time, at random, until only balls of one colour are left. Show that they are all white with probability $n/(n+m)$.

Solution.

Step 1. Continue the withdrawals until the urn is empty; the $(n+m)!$ orders of labelled balls are equally likely, and the experiment is an initial part of the order.

Step 2. The balls left form a nonempty final segment (both colours are present at the start), containing the last ball, and all have its colour. So they are white iff the last ball is white.

Step 3. Each ball is last in $(n+m-1)!$ orders, i.e. with probability $1/(n+m)$; so $P = n/(n+m)$. ■

Exercise 3.16 (An even number of successes)

Let P_n be the probability that n Bernoulli trials with success probability p result in an even number of successes (0 counts as even). Show that

$$P_n = p(1-P_{n-1}) + (1-p)P_{n-1} \quad (n \geq 1),$$

and deduce $P_n = (1+(1-2p)^n)/2$.

Solution.

Step 1. With S the successes in the first $n-1$ trials, the total is even iff trial n succeeds and S is odd, or fails and S is even; by independence

$$P_n = p\,P(S \text{ odd}) + (1-p)P(S \text{ even}) = p(1-P_{n-1}) + (1-p)P_{n-1}.$$

Step 2. I.e. $P_n = p + (1-2p)P_{n-1}$, with $P_0 = 1$. Inductively,

$$P_n = p + \frac{1-2p}{2} + \frac{(1-2p)^n}{2} = \frac{1+(1-2p)^n}{2},$$

■

Exercise 3.18 (No run of three heads)

Let Q_n be the probability that no run of three consecutive heads appears in n tosses of a fair coin. Show that

$$Q_n = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3} \quad (n \geq 3), \qquad Q_0 = Q_1 = Q_2 = 1,$$

and find Q_8.

Solution.

Step 1. Let R: no run of three heads; F_j: first tail at toss j ($j = 1, 2, 3$); H: tosses $1, 2, 3$ all heads. These partition the space, $P(F_j) = 2^{-j}$.

Step 2. $RH = \emptyset$. On F_j the first j tosses contain no run and toss j is a tail, so $RF_j = F_j \cap R_j$, R_j: no run among tosses $j+1, \ldots, n$.

Step 3. R_j is independent of F_j and $P(R_j) = Q_{n-j}$:

$$Q_n = \sum_{j=1}^{3} P(F_j)P(R_j) = \tfrac{1}{2}Q_{n-1} + \tfrac{1}{4}Q_{n-2} + \tfrac{1}{8}Q_{n-3}.$$

Step 4. $Q_3 = \frac{7}{8}$, $Q_4 = \frac{13}{16}$, $Q_5 = \frac{24}{32}$, $Q_6 = \frac{44}{64}$, $Q_7 = \frac{81}{128}$ and

$$Q_8 = \tfrac{1}{2}\cdot\tfrac{81}{128} + \tfrac{1}{4}\cdot\tfrac{44}{64} + \tfrac{1}{8}\cdot\tfrac{24}{32} = \frac{81+44+24}{256} = \frac{149}{256} \approx 0.582.$$

■

Exercise 3.21 (The ballot problem)

In an election, candidate A receives n votes and candidate B receives m votes, $n > m \geq 0$, and all $\binom{n+m}{n}$ orderings of the votes are equally likely. Let $P_{n,m}$ be the probability that A is strictly ahead throughout the count.
(a) Derive a recursion for $P_{n,m}$ in terms of $P_{n-1,m}$ and $P_{n,m-1}$ by conditioning on who receives the last vote.
(b) Prove that $P_{n,m} = (n-m)/(n+m)$.

Solution.

(a)

Step 1. Orderings ending with A correspond to orderings of $n-1$ A's and m B's, so the last vote is A's with probability $\binom{n+m-1}{n-1}/\binom{n+m}{n} = n/(n+m)$, and then the first $n+m-1$ votes are a uniform ordering of $n-1$ A's and m B's; likewise for B.

Step 2. Since $n > m$, A is ahead at the end, so A leads throughout iff A leads throughout the first $n+m-1$ votes. For $m \geq 1$

$$P_{n,m} = \frac{n}{n+m} P_{n-1,m} + \frac{m}{n+m} P_{n,m-1},$$

with $P_{k,k} = 0$ ($k \geq 1$; a tie at the end) and $P_{n,0} = 1$.

(b)

Step 1. Induction on $n+m$. $m = 0$: $P_{n,0} = 1 = n/n$; the formula gives 0 when $n = m$, matching $P_{k,k} = 0$.

Step 2. For $n > m \geq 1$, by (a) and the hypothesis (covering $P_{n-1,m}$ also when $n - 1 = m$),

$$\begin{aligned} P_{n,m} &= \frac{n}{n+m} \cdot \frac{n-1-m}{n+m-1} + \frac{m}{n+m} \cdot \frac{n-m+1}{n+m-1} \\ &= \frac{(n-m)(n+m) - (n-m)}{(n+m)(n+m-1)} = \frac{n-m}{n+m}, \end{aligned}$$

using $n(n-m-1) + m(n-m+1) = n^2 - m^2 - (n-m)$. ■